

peak and mean powers both calculated correctly, with supporting working, and compared leading to conclusion that mean power is half the peak power **A1**

8 (a) (i) $a = \omega^2 r = \left(\frac{2\pi}{24 \times 60 \times 60}\right)^2 (6370 \times 10^3)$ **C1**
 $= 3.37 \times 10^{-2} \text{ ms}^{-2}$ **A1**

(ii) $F_G = \frac{GMm}{r^2}$
 $= \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})(1.00)}{(6370 \times 10^3)^2}$ **C1**
 $= 9.83 \text{ N}$ **A1**



B1: Correct Magnitude
 B1: Correct Direction

(ii) $F = ma$
 $F_G - F_s = ma$

$$9.83 - F_s = 1.00 \times 3.37 \times 10^{-2} \quad \text{C1}$$

$$F_s = 9.80 \text{ N} \quad \text{A1}$$

- (iii) There is a centripetal force acting on the ball. B1
 The measured acceleration would be slightly less than the gravitational acceleration alone. B1

- (c) (i) When the tension in the rope is more than the weight of the person, the resultant force acting on the person is opposite in direction to his velocity and the person is decelerating. When the tension in the rope is less than the weight of the person, the resultant force acting on the person is in the same direction as his velocity and the person is accelerating. B1
 Thus, the person will have maximum speed when the resultant force acting on him is zero and with a corresponding acceleration of zero. B1

- (ii) Man is moving at maximum speed when resultant force acting on him is zero.

$$mg = ke$$

$$e = \frac{mg}{k} = \frac{80.0 \times 9.81}{120} = 6.54 \text{ m} \quad \text{C1}$$

Applying Principle of Conservation of Energy,

Total energy before stepping off = Total energy at maximum speed

$$0 + 0 + 0 = \frac{1}{2}mv^2 + \frac{1}{2}ke^2 - mg(l + e) \quad \text{C1}$$

$$v = \sqrt{\frac{2mg(l+e) - ke^2}{m}}$$

$$= \sqrt{\frac{2 \times 80.0 \times 9.81 \times (25.0 + 6.54) - 120 \times 6.54^2}{80.0}}$$

$$= 23.6 \text{ m s}^{-1} \quad \text{A1}$$

- (iii) Applying Principle of Conservation of Energy,
 GPE at highest point = EPE at lowest point

$$mg(l + e) = \frac{1}{2}ke^2 \quad \text{C1}$$

$$e = 25.8 \text{ m} \quad \text{A1}$$

- (iv) 1. At highest point,
 $GPE = mg(l + e) = 80.0 \times 9.81 \times (25.0 + 25.8)$
 $= 3.99 \times 10^4 \text{ J} \quad \text{C0}$

Straight line with correct labels on y-axis and x-axis B1

2. Exponential curve with correct start and end points B1

$$3. \quad E_{k0} = \frac{1}{2}mv_0^2 = \frac{1}{2} \times 80.0 \times 23.6^2 = 2.23 \times 10^4 \text{ J}$$

C0

Straight line from $s = 0 \text{ m}$ to $s = 25.0 \text{ m}$

B1

Curve from $s = 25.0 \text{ m}$ to $s = 50.8 \text{ m}$, with correct label on y-axis

